



PES
UNIVERSITY
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B. Tech – II

Department of Science and
Humanities

CLASS-3

General property of LAPLACE TRANSFORM

Change of scale

Time scaling. Let $\mathcal{L}\{f(t)\} = F(s)$ then

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right),$$

for any $a > 0$.

To prove, we start again with the definition

$$\mathcal{L}\{f(at)\} = \int_0^{\infty} f(at)e^{-st} dt,$$

and use the change of variables $at = \tau$, from where $dt = d\tau/a$. Note that for $a > 0$ the limits of integration will not change:

$$\mathcal{L}\{f(\tau)\} = \frac{1}{a} \int_0^{\infty} f(\tau)e^{-\frac{s}{a}\tau} d\tau = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Example

Find $\mathcal{L} \{\cos 3t\}$.

By the previous property and the fact that $\mathcal{L} \{\cos t\} = \frac{s}{s^2+1}$

$$\begin{aligned}\mathcal{L} \{\cos 3t\} &= \frac{1}{3} \frac{s/3}{(s/3)^2 + 1} \cdot \\ &= \frac{s}{s^2 + 3^2} \cdot\end{aligned}$$

First shift property

Shifting property. If $\mathcal{L}\{f\} = F(s)$ then $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$.

To prove this property, consider

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^{\infty} e^{at}f(t)e^{-st} dt \\ &= \int_0^{\infty} e^{-(s-a)t}f(t) dt \quad (\text{let } p = s - a) \\ &= \int_0^{\infty} e^{-pt}f(t) dt = F(p) = F(s - a).\end{aligned}$$

Example

$$\mathcal{L} \{e^{3t} \sin t\}$$

$$\text{since } \mathcal{L} \{\sin t\} = \frac{1}{s^2+1}.$$

$$\mathcal{L} \{e^{at} f(t)\} = F(s-a).$$

$$\mathcal{L} \{e^{3t} \sin t\} = \frac{1}{(s-3)^2+1},$$

Differentiation of transforms/ multiplication by t



Differentiation of the frequency.

$$\text{Let } \mathcal{L} \{f(t)\} = F(s).$$

$$\text{Then } \mathcal{L} \{tf(t)\} = -F'(s).$$

Or, more generally,

$$\mathcal{L} \{t^n f(t)\} = (-1)^n F^{(n)}(s), \quad n \in \mathbf{N}.$$

To show that this is true, consider the derivative of $F(s)$:

$$F'(s) = \int_0^{\infty} (-t)f(t)e^{-st} dt,$$

Example $\mathcal{L}\{t \cos 7t\}$

$$\mathcal{L}\{tf(t)\} = -F'(s) = -\left(\frac{d}{ds}F(s)\right),$$

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 7t\} = \frac{s}{s^2 + 7^2}$$

So

$$\frac{d}{ds}F(s) = \frac{d}{ds} \frac{s}{s^2 + 7^2} = -\frac{s^2 - 7^2}{(s^2 + 7^2)^2}$$

Example

$$L(t^2 e^{at})$$

$$\text{Sol: } -L(t^2 e^{at}) = (-1)^2 \frac{d^2}{ds^2} \left(\frac{1}{s-a} \right)$$

$$= \frac{d}{ds} \left(\frac{-1}{(s-a)^2} \right)$$

$$= \frac{2}{(s-a)^3}$$



THANK YOU

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